

Data Flow Diagram

Date	30 September 2026
Team ID	XXXXXX
Project Name	Learnmate – AI StudyBuddy
Maximum Marks	2 Marks

Data Flow Diagram

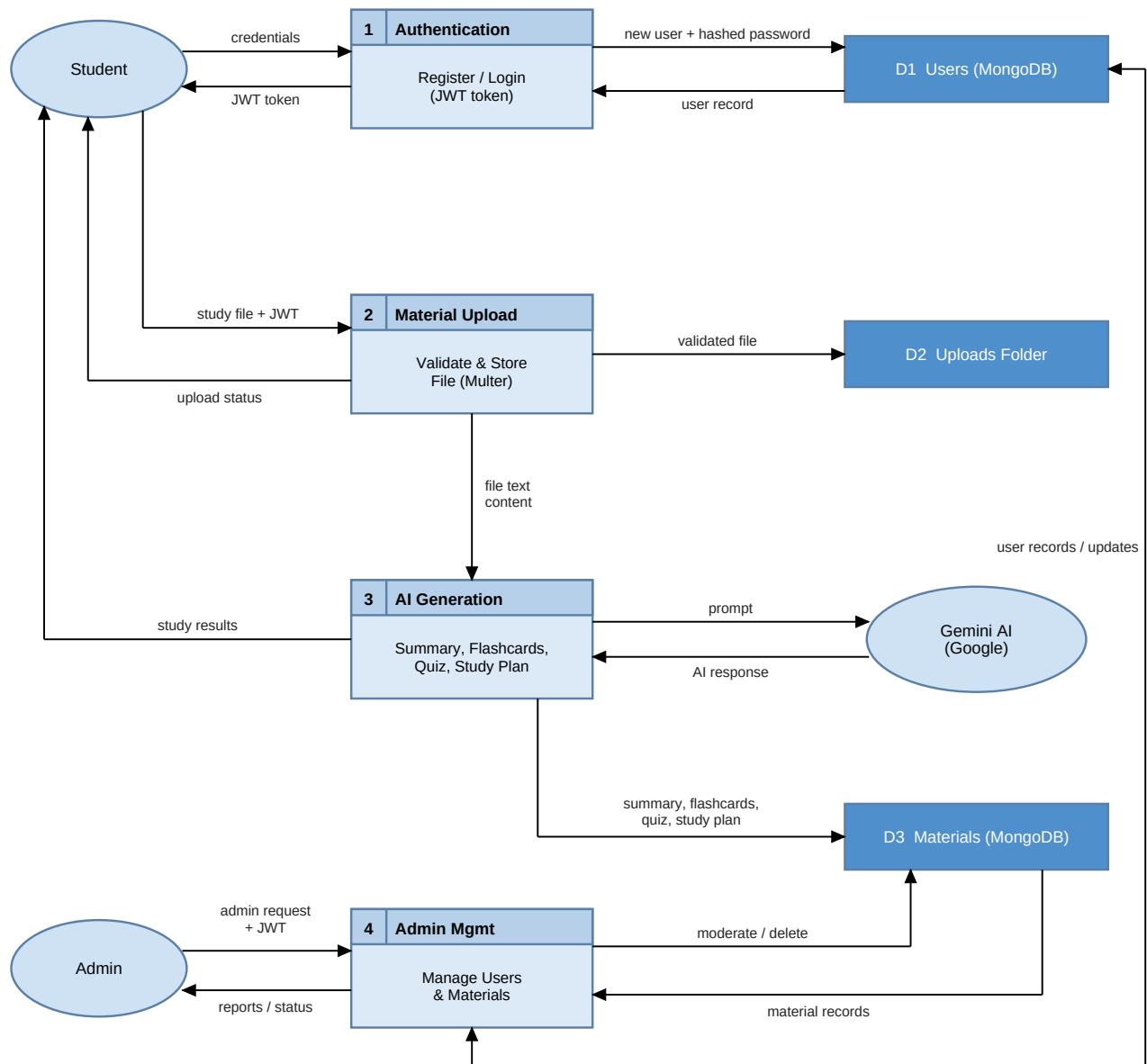
A Data Flow Diagram (DFD) shows how data moves through Learnmate – between external entities, processes, and data stores. The symbol legend below is used in the diagram.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Student, Admin, Gemini AI).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Authentication, AI Generation).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Users and Materials collections in MongoDB).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Learnmate Data Flow Diagram (Level 1)

The diagram below shows how a student registers, uploads study material, and receives AI-generated study content, and how an admin manages users and materials.



Processes

No.	Process	Description
1	Authentication	Registers users (password hashed with bcrypt), verifies login, and issues a JWT access token. Files: authController.js, auth.js, tokens.js.
2	Material Upload	Accepts .txt, .md and .pdf files up to 5 MB via Multer and saves them to the uploads folder. Files: upload.js, materialController.js.
3	AI Generation	Sends the material text to Gemini (with retry on 429/503 errors) and saves the summary, flashcards, quiz and study plan. File: gemini.js.
4	Admin Management	Admin-only routes (role check) to view and manage users and materials. Files: adminController.js, admin.js.

Data Stores and External Entities

ID / Name	Type	Contents
D1 – Users	Data Store (MongoDB)	name, email, hashed password, role (student / admin), timestamps.
D2 – Uploads Folder	Data Store (server files)	Uploaded study files with unique generated file names.
D3 – Materials	Data Store (MongoDB)	user, title, content, filename, summary, flashcards, quiz, study plan.
Student	External Entity	Registers, uploads material, and studies using the AI-generated content.
Admin	External Entity	Monitors and manages users and study materials.
Gemini AI (Google)	External Entity	Receives prompts and returns generated study content.